SIMATS ENGINEERING

CSA1511 – CLOUD COMPUTING AND BIG DATA ANALYTICS

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Title: Cloud Computing Principles in an Online Bus Booking System

1. Introduction

An Online Bus Booking System is a web and mobile application that allows users to search buses, check seat availability, book tickets, make online payments, and receive booking confirmations.

Cloud computing enables the application to handle thousands of users simultaneously without slowing down. It also stores booking data securely and provides uninterrupted service.

Examples:

- RedBus
- AbhiBus
- MakeMyTrip Bus Booking

2. Cloud Computing Principles Used

A. On-Demand Self-Service

Users can:

- Search buses
- Book tickets
- Cancel bookings
- Download tickets

without contacting customer support.

Example:
A passenger books a Chennai to Bangalore ticket anytime using the app.

B. Broad Network Access

The application is accessible through:

- Mobile phones
- Laptops
- Tablets
- Desktop computers

using the Internet.

C. Resource Pooling

The cloud shares computing resources among millions of users.

Example:

- Booking database
- Web servers
- Payment services

are shared efficiently without affecting user experience.

D. Rapid Elasticity (Scalability)

During holidays or festivals:

- Thousands of people book tickets at the same time.

Cloud automatically:

- Adds more servers
- Increases processing power
- Balances user traffic

When traffic decreases, extra servers are removed automatically.

E. Measured Service

Cloud providers monitor:

- CPU usage
- Storage usage
- Network bandwidth

- Database usage

The company pays only for the resources it actually uses.

3. Cloud Service Models

Service Model	Usage in Bus Booking
IaaS	Virtual machines, storage, networking
PaaS	Hosting backend APIs and databases
SaaS	Users access the booking application through a browser or mobile app

4. Cloud Deployment Model

The Bus Booking System mainly uses:

Public Cloud

because it provides:

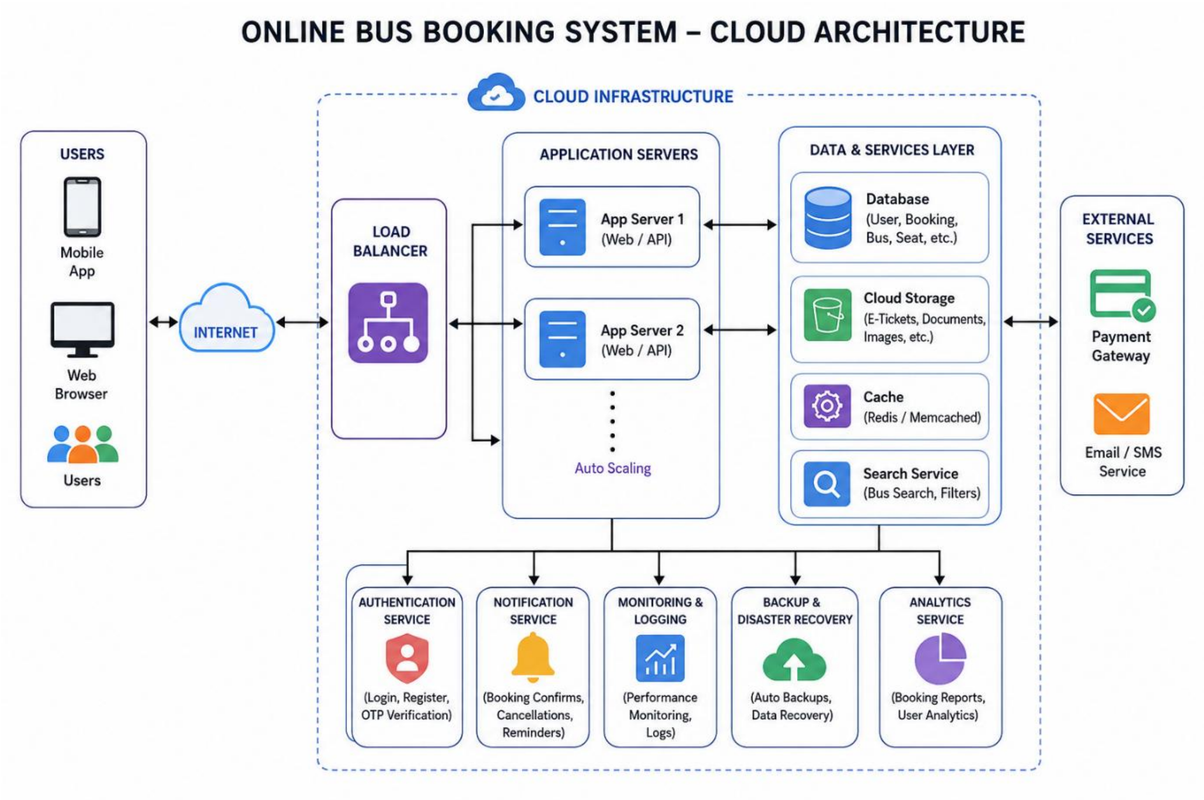
- High availability
- Scalability
- Cost efficiency
- Global access

Examples:

- AWS

- Microsoft Azure
- Google Cloud Platform

5. Architecture



6. Cloud Components Used

Frontend

- Website
- Mobile App

Backend

- Booking API

- Authentication
- Seat Management

Database

Stores:

- User details
- Booking history
- Bus schedules
- Seat availability

Cloud Storage

Stores:

- E-ticket PDF
- User documents
- Images

Notification Service

Sends:

- Booking confirmation
- Cancellation message
- Payment receipt

7.Features Enabled by Cloud Computing

- Online registration
- Login authentication

- Bus search
- Seat selection
- Secure payment
- Ticket generation
- Booking history
- Email notifications
- SMS notifications
- 24×7 availability

8.Cloud Security

The application uses cloud security to protect user information.

Security measures include:

- User authentication
- Password encryption
- HTTPS communication
- Secure payment gateway
- Database backup
- Firewall protection

Advantages of Using Cloud Computing

- Fast booking process

- High availability
- Automatic scaling
- Secure data storage
- Easy maintenance
- Cost-effective
- Backup and disaster recovery
- Global accessibility

9.Real-Life Scenario

Suppose it is Diwali.

Around **5 lakh users** open the application at the same time.

Without cloud computing:

- Website becomes slow
- Server crashes
- Booking fails

With cloud computing:

- New servers are automatically created.
- User traffic is distributed using a load balancer.
- Booking continues smoothly.
- After the rush, unnecessary servers are removed to reduce cost.

This demonstrates the cloud principle of **Rapid Elasticity**.

Advantages to Users

- Easy online booking
- Instant confirmation
- Secure online payment
- Ticket download
- Booking cancellation
- Refund tracking
- Access from anywhere

10.Technologies Commonly Used

- Frontend: HTML, CSS, JavaScript, Android/iOS
- Backend: Node.js, Java, Python, .NET
- Database: MySQL, PostgreSQL, MongoDB
- Cloud Platform: AWS, Azure, Google Cloud
- Payment Gateway: Razorpay, Stripe, PayPal
- Email Service: SMTP or cloud email services

11.Conclusion

Cloud computing plays a vital role in an Online Bus Booking System by providing scalable infrastructure, secure data storage, high availability, and reliable performance. It enables users to book tickets anytime and from anywhere while allowing the service provider to efficiently manage large volumes of bookings. By using cloud principles such as on-demand self-service, resource pooling, rapid elasticity, broad network access, and measured service, the application delivers a seamless and dependable booking experience.

12.Result

The activity was completed successfully by analyzing an Online Bus Booking System and understanding how cloud computing principles such as on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service are applied to provide a secure, scalable, and highly available online booking platform.